

Lesson 6: Problem Solving with Triangles

Saturday, August 10, 2024 6:55 PM

Lesson 6: Problem Solving with Triangles

The Pythagorean Theorem ✓

Find Missing Angles ~~_____~~ we already did this

Missing Sides in Similar Figures ✓

Indirect Measurement ✓



common:
a 90° angle

THE PYTHAGOREAN THEOREM

A right triangle has a right angle, which equals 90°. The two sides that form the right angle are called **legs**, and the side opposite the right angle is the **hypotenuse**. The hypotenuse is always the longest side of a right triangle.

The relationship between the legs and hypotenuse of a right triangle was discovered almost 2,500 years ago by a Greek mathematician named Pythagoras.

Pythagorean theorem $a^2 + b^2 = c^2$

The sum of the squares of the legs of a right triangle equals the square of the hypotenuse.

FINDING THE HYPOTENUSE

Example The lengths of the legs of a right triangle are 9 and 12. What is the hypotenuse?

Step 1

Substitute 9 for a and 12 for b in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$

$$9^2 + 12^2 = c^2$$

The hypotenuse is 15 units long.

Step 2

Evaluate the left side of the equation.

$$\sqrt{81 + 144} = c$$

$$\sqrt{225} = c$$

Step 3

Find c .

$$15 = c$$

$$a^2 + b^2 = c^2$$

$$c = \sqrt{a^2 + b^2}$$

$$= \sqrt{9^2 + 12^2}$$

$$= \sqrt{81 + 144}$$

$$= \sqrt{225}$$

$$a^2 + b^2 = c^2$$

$$(\text{leg } 1)^2 + (\text{leg } 2)^2 = (\text{hyp})^2$$

$$c = 15$$

Sosan Amiri 9:44 AM

The two sides that form the right angle are called legs, and the side opposite the right angle is the hypotenuse. right?

Soraiya:



FINDING THE LEG OF A RIGHT TRIANGLE

Example The hypotenuse of a right triangle has a length of 10 units, and one of the legs is 6 units long. Find the measure of the other leg.

Step 1

Substitute 10 for c and 6 for a in the Pythagorean theorem formula.

$$a^2 + b^2 = c^2$$

$$6^2 + b^2 = 10^2$$

The other leg is 8 units long.

Step 2

Simplify the equation.

$$36 + b^2 = 100$$

$$b = \sqrt{64}$$

Step 3

Find b .

$$b = 8$$

$$a^2 + b^2 = c^2$$

$$a^2 = c^2 - b^2$$

$$a = \sqrt{c^2 - b^2}$$

$$= \sqrt{10^2 - 6^2}$$

$$= \sqrt{100 - 36}$$

$$= \sqrt{64}$$

$$a = 8$$

$$b = \sqrt{c^2 - a^2}$$

WORDS

Rt. Δ leg & hyp
other Δ base & height

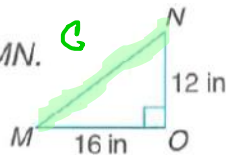
Naima Rahimi 9:45 AM

And the Hypotenuse is always the longest in right triangle

Pythagorean Theorem - Examples Set #1

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1. Find the measurement of side MN .



$$a^2 + b^2 = c^2$$

$$c = \sqrt{a^2 + b^2}$$

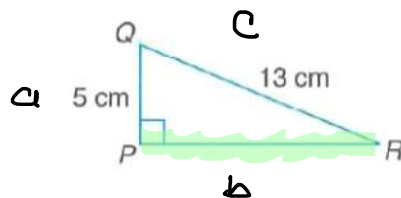
$$= \sqrt{16^2 + 12^2}$$

$$= \sqrt{256 + 144}$$

$$= \sqrt{400}$$

$$\boxed{c = 20 \text{ in}}$$

2. What is the measurement of side PR ?



$$a^2 + b^2 = c^2$$

$$b^2 = c^2 - a^2$$

$$\boxed{b = \sqrt{c^2 - a^2}}$$

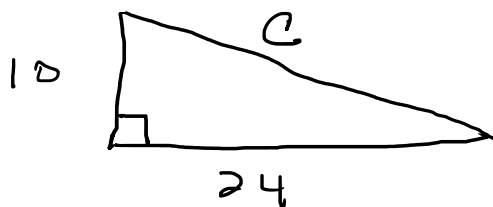
$$b = \sqrt{13^2 - 5^2}$$

$$= \sqrt{169 - 25}$$

$$= \sqrt{144}$$

$$\boxed{b = 12 \text{ cm}}$$

3. The legs of a right triangle measure 10 inches and 24 inches. What is the hypotenuse?



$$c = \sqrt{a^2 + b^2}$$

$$= \sqrt{10^2 + 24^2}$$

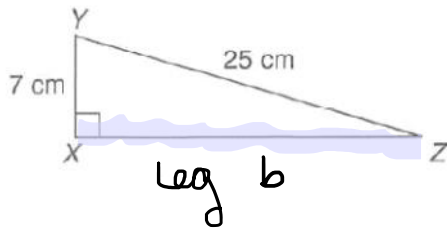
$$= \sqrt{100 + 576}$$

$$= \sqrt{676}$$

$$\boxed{c = 26}$$

4. What is the measurement of side XZ ?

4. What is the measurement of side XZ?



$$a^2 + b^2 = c^2$$

$$b = \sqrt{c^2 - a^2}$$

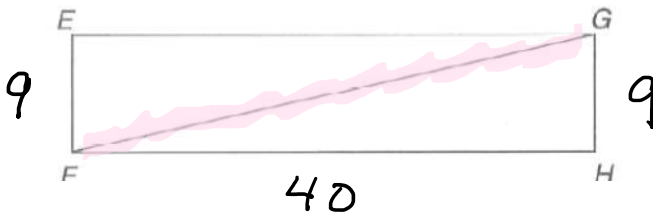
$$= \sqrt{25^2 - 7^2}$$

$$= \sqrt{625 - 49}$$

$$= \sqrt{576}$$

$$b = 24 \text{ cm}$$

5. In the rectangle, side EF is 9 cm and side FH is 40 cm. Find the measure of the diagonal FG.



$$a^2 + b^2 = c^2$$

$$9^2 + 40^2 = c^2$$

$$81 + 1600 = c^2$$

$$c^2 = 1681$$

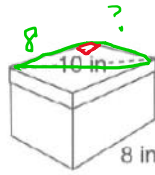
$$c = \sqrt{1681}$$

$$c = 41 \text{ cm}$$

Pythagorean Theorem - Examples Set #2

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6. If the lid of a box has a diagonal length of 10 inches and a length of 8 inches, what is the width of the box in inches?



$$8^2 + b^2 = 10^2$$

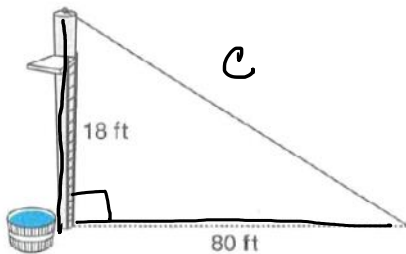
$$b = \sqrt{10^2 - 8^2}$$

$$= \sqrt{100 - 64}$$

$$= \sqrt{36}$$

$$b = 6 \text{ in}$$

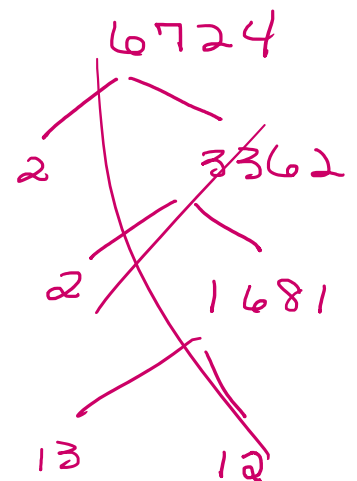
7. For a circus act, an 18-foot pole is braced with a wire that extends from the top of the pole to a stake in the ground 80 feet from the base of the pole. How long is the bracing wire?



$$\begin{aligned} C &= \sqrt{a^2 + b^2} \\ &= \sqrt{18^2 + 80^2} \\ &= \sqrt{324 + 6400} \\ &= \sqrt{6724} \end{aligned}$$

$$C = 82'$$

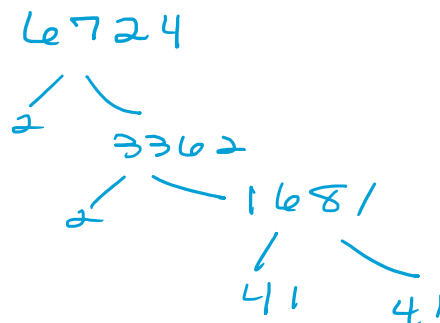
PRIME
FACTORIZATION
TREE



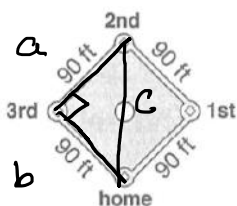
give up

$$\begin{array}{r} 3362 \\ 2 \overline{) 6724} \\ \underline{6} \\ 7 \\ \underline{6} \\ 12 \\ \underline{12} \\ 0 \end{array}$$

$$\begin{array}{r} 1681 \\ 2 \overline{) 3362} \\ \underline{2} \\ 18 \\ \underline{12} \\ 16 \\ \underline{16} \\ 0 \end{array}$$



8. The base paths on a baseball diamond form right angles. To the nearest tenth of a foot, what is the distance from home plate to second base?



$$162 = 2(9^2)$$

$$2 \quad \swarrow \quad \searrow$$

$$81$$

$$\swarrow \quad \searrow$$

$$9 \quad 9$$

$$\sqrt{2} \approx 1.4$$

$$90 \times 1.5 = 135$$

$$c = \sqrt{a^2 + b^2}$$

$$= \sqrt{90^2 + 90^2}$$

$$= \sqrt{8100 + 8100}$$

$$= \sqrt{16200} \checkmark$$

$$= \sqrt{162(10^2)}$$

$$= 10\sqrt{162}$$

$$= 10\sqrt{2(9^2)}$$

$$= 90\sqrt{2} \checkmark$$

$$c = 127.3 \text{ ft} \checkmark$$

Summary

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A set of whole numbers that forms a right triangle is called a **Pythagorean triple**.

<i>a</i>	<i>b</i>	<i>c</i>
3	4	5
6	8	10
9	12	15

Finding Missing Angles

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This is a repeat of what we learned in Lesson 1, Angles.

SIMILAR GEOMETRIC FIGURES

Similar figures have the same shape but are different sizes. Imagine using a copy machine to reduce a figure you have drawn. The figure on the copy would be in a smaller scale than the original, but the shape would be the same.

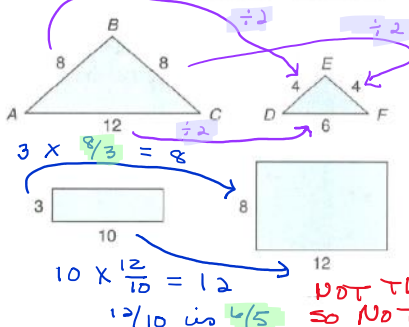
These are proportional to each other.

The triangles to the right are similar because the ratios of the lengths of corresponding sides are equal. Each side in the second triangle is half as long as the first.

$ABC \sim DEF$ The symbol $\sim$ means "is similar to."

The rectangles shown are not similar.

You can use proportion to solve problems about similar figures.



FINDING A MISSING SIDE

Example The plots of land shown here are similar. Find the width of the smaller plot.

Step 1

Set up a proportion with corresponding sides.

$$\frac{\text{length}}{\text{width}} = \frac{50}{40} = \frac{30}{x}$$

The width of the smaller plot is 24.

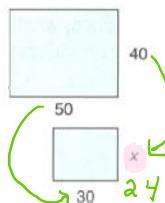
Step 2

Cross multiply and solve for x .

$$50x = 1,200$$

$$x = 24$$

$$50 \times \frac{3}{5} = 30$$



a different way to solve
since we mult 50 by $\frac{3}{5}$
to get 30, let's do the
same to 40:
 $40 \times \frac{3}{5} = 24$

Solve for x so that the
figures are similar.

50 is to 30 as 40 is to x

$$50 : 30 = 40 : x$$

$$50/30 = 40/x$$

$$(x) 5/3 = 40/x (x)$$

$$(\frac{3}{5}) 5x/3 = 40 (\frac{3}{5})$$

$$x = 120/5$$

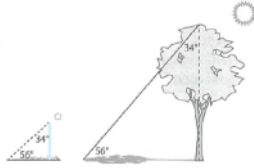
$$x = 24$$

Similar Triangles - Examples Set #1

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- ② Their sides are proportional to each other

① Two triangles are similar if their corresponding angles are equal. When the sun casts the shadow of an object that is perpendicular to the ground, a right triangle is formed. The angle of the sun will be the same for any two objects, creating equal corresponding angles. You can find the height of a tall object by using a smaller object and comparing similar triangles.



$$\frac{4}{x} = \frac{4}{10} \rightarrow \left(\frac{5x}{2}\right) \left(\frac{1}{x}\right) = \left(\frac{2x}{5}\right) \left(\frac{5}{2}\right)$$

$$\frac{4}{x} = \frac{2}{5} \rightarrow 15 = x$$

we could:

4 is to 4 as x is to 10

$$6:4 = x:10$$

$$\frac{6}{4} = \frac{x}{10}$$

$$(10) \frac{3}{2} = \frac{x}{10} (10)$$

$$15 = x$$

$$x = 15$$

FINDING THE HEIGHT OF A TALL OBJECT

Example A street sign 6 feet high casts a shadow of 4 feet. At the same time, a nearby tree casts a 10-foot shadow. How tall is the tree?

Step 1

Set up a proportion with corresponding sides.

$$\frac{\text{height}}{\text{shadow}} = \frac{6}{4} = \frac{x}{10}$$

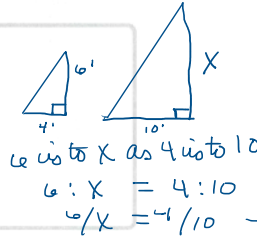
The height of the tree is 15 feet.

Step 2

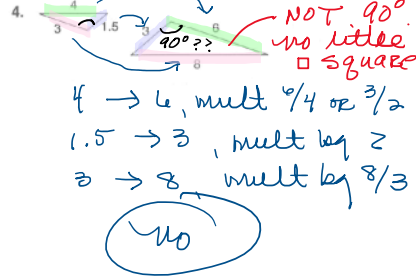
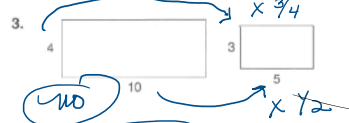
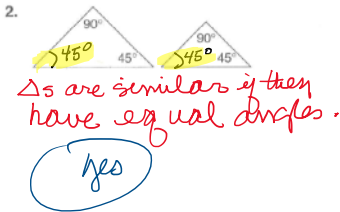
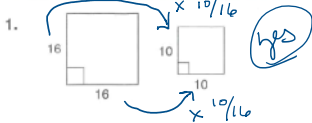
Cross multiply and solve for x.

$$4x = 60$$

$$x = 15$$



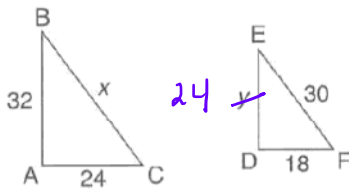
Decide whether each pair of figures is similar. Do not rely on the appearance of the drawings.



Similar Triangles - Examples Set #2

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5. Triangles ABC and DEF are similar. Find the missing lengths.



$$\begin{aligned} 32 \text{ is to } 24 \text{ as } y \text{ is to } 18 \\ 32:24 &= y:18 \\ 32/24 &= y/18 \\ (18) 4/3 &= y/18 \quad (18) \end{aligned}$$

$$\begin{aligned} 24 &= y \\ y &= 24 \end{aligned}$$

$$\begin{aligned} 32 \text{ is to } x \text{ as } 24 \text{ is to } 30 \\ 32:x &= 24:30 \\ 32/x &= 24/30 \end{aligned}$$

$$\begin{aligned} (x) 32/x &= 4/5 (x) \\ (5/4) 32 &= 4x/5 (5/4) \end{aligned}$$

$$\begin{aligned} 40 &= x \\ x &= 40 \end{aligned}$$

6. Karlene is using a photocopier to enlarge a graph that is 3 inches wide and 5 inches long. If her boss wants the length of the enlargement to be $7\frac{1}{2}$ inches, how wide will the graph be?



$$\begin{aligned} 7\frac{1}{2} \\ \text{this is } 7 \text{ plus } \frac{1}{2} \\ &= 7.5 \\ &= 15/2 \end{aligned}$$

✓ 3 is to 5 as x is to $7\frac{1}{2}$

✓ $3:5 = x:7\frac{1}{2}$

✓ $3/5 = x/7\frac{1}{2}$

✓ $3/5 = x/(15/2)$

✓ $(15/2) 3/5 = 2x/15 (15/2)$

✓ $9/2 = x$

$$x = 4.5$$

$$\begin{aligned} x &\div 15/2 \\ &= x (2/15) \end{aligned}$$

7. A billboard casts a shadow of 39 feet when a nearby 4-foot post casts a shadow of 6 feet.

Reliable
Used Cars

3 is to 2 as 39 is to x ✓

7. A billboard casts a shadow of 39 feet when a nearby 4-foot post casts a shadow of 6 feet. Find the height of the billboard.



3 is shadow

2 is height

39 is shadow

? is height

3 is to 2 as 39 is to x ✓

$$3 \cdot 2 = 39 \cdot x$$

$$(x) \quad 3/2 = 39/x \quad (x)$$

$$(2/3) \quad 3x/2 = 39 \quad (2/3)$$

$$x = 78/3$$

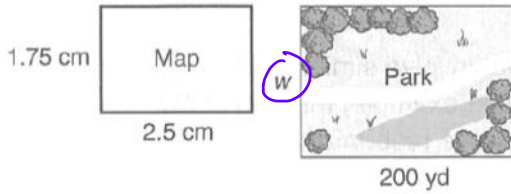
$$x = 26 \text{ ft}$$

in step

Similar Triangles - Examples Set #3

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8. On a map, a rectangular park is 1.75 centimeters wide by 2.5 centimeters long. If the park is actually 200 yards long, what is its width in yards?



$$\frac{w}{1.75} \text{ is to } \frac{L}{2.5} \text{ as } \frac{w}{w} \text{ is to } \frac{L}{200}$$

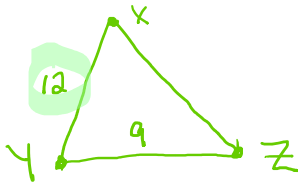
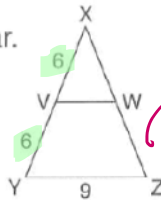
$$1.75 : 2.5 = w : 200$$

$$(200) \cdot 1.75 / 2.5 = w / 200 \quad (200)$$

$$140 = w$$

$$w = 140 \text{ yds}$$

9. Triangles XYZ and XVW are similar. What is the length of VW?



$$\checkmark 6 \text{ is to } \overline{VW} \text{ as } 12 \text{ is to } 9$$

$$\checkmark 6 : \overline{VW} = 12 : 9$$

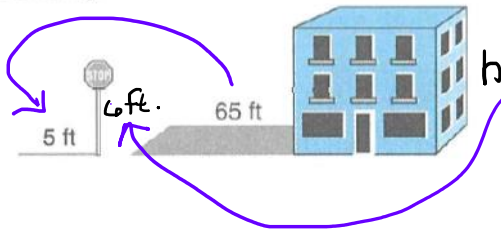
$$(\frac{9}{12}) \overline{VW} / 6 / \overline{VW} = 12 / 9 \quad (\frac{9}{12}) \overline{VW}$$

$$(\frac{9}{12}) (6) = \overline{VW}$$

$$4.5 = \overline{VW}$$

$$\overline{VW} = 4.5$$

10. A building casts a shadow of 65 feet while a nearby street sign casts a shadow of 5 feet. If the street sign is 6 feet in height, how high is the building?



$$\frac{h}{65} \text{ is to } \frac{L}{5} \text{ as } \frac{h}{h} \text{ is to } \frac{L}{65}$$

$$\text{OR} \quad \frac{L_1}{5} \text{ is to } \frac{h_1}{6} \text{ as } \frac{L_2}{65} \text{ is to } \frac{h_2}{h}$$

$$\text{OR} \quad \frac{L_1}{5} \text{ is to } \frac{L_2}{65} \text{ as } \frac{h_1}{6} \text{ is to } \frac{h_2}{h}$$

$$\text{OR} \quad \frac{65 \text{ is to } 5 \text{ as } h \text{ is to } 6}{L_2 \quad L_1 \quad h_2 \quad h_1}$$

$$\frac{L_1}{5} \text{ is to } \frac{h_2}{h} \text{ as } \frac{L_2}{65} \text{ is to } \frac{h_1}{6}$$

can we do it this way?

No, order of L s & h s are inconsistent
not correct

$$65 : 5 = h : 6$$

$$65 = h / 6$$

$$(6) 13 = h / 6 \quad (6)$$

$$78 = h$$

$$h = 78 \text{ ft}$$

always write

$$h_2 : L_1 = h_1 : L_2$$

~~always true~~

$$h_2 : L_1 = h_1 : L_2$$

The variable = number

$$h = 78$$

~~never $78 = h$~~

are inconsistent
NOT CONSISTENT.

Also BONG

h_1 to h_2 as L_2 to L_1

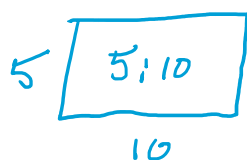
Similar Triangles - Examples Set #4

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11. **Analyze** Write *True* or *False* after the following statements. Write one sentence explaining your reasoning for each one.

- All equilateral triangles are similar.
- All squares are similar.
- All rectangles have equal angles; therefore, all rectangles are similar.
- All right triangles are similar.

c) False. Counter-Example



5:10 is $\frac{1}{2}$



1:18 is $\frac{1}{18}$

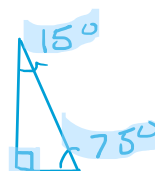
a) TRUE

all angles are 60°
so every equilateral Δ
has three 60° angles

b) True

the Ratio of W to L
of all squares is 1:1

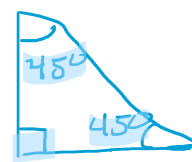
d) False. Counter-Examples



75:15

Ratio is
5 to 1

or just 5



45:45

Ratio is
1 to 1

or just 1



FINDING PATTERNS IN ALGEBRA AND GEOMETRY

Finding a pattern is a useful problem-solving strategy. It can save time you might otherwise spend in lengthy calculations. In this lesson, you will see how patterns can be helpful in solving problems.

One example of patterns in mathematics is the **number series**. A number series is a set of numbers that continues according to a rule.

The counting numbers (1, 2, 3, ...) are a number series in which each number is 1 larger than the number before it. In the series 1, 4, 7, 10, ... each number is 3 more than the preceding number.

The Rule is ADD 1

Rule is ADD 3 ... 3 dots are called ELLIPSES

and it means ET CETERA (Latin for and so on)

Find the next number in each of the following series. Write the rule that determines which number comes next.

1. 3, 6, 9, 12, 15, Add 3, 18

2. 100, 91, 82, 73, 64, subtract 9, 55

3. 1, 10, 100, 1,000, Multiply 10

10⁰, 10¹, 10², 10³ Add 1 to the exponent, 10⁴

Complete each series.



7. 1, 5, 9, 13, 17 ADD 4, U

8. A, E, I, M, Q, _____
(Hint: Think about the position of each letter in the alphabet.)

9. 1, 4, 9, 16, 25, _____

10. 9 in, 1 ft 6 in, 2 ft 3 in, 3 ft, 3 ft 9 in, _____

11. -2, 4, -8, 16, -32, _____

4. 2, 4, 8, 16, Multiply 2, 32

5. 2, 5, 11, 23, 47, 95

6. 1, $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, divide 2, $\frac{1}{16}$ multiply 2, $\frac{1}{16}$

#5

$$2 = 3 - 1 = 1(3) - 1 = 2^0(3) - 1$$

$$5 = 6 - 1 = 2(3) - 1 = 2^1(3) - 1$$

$$11 = 12 - 1 = 4(3) - 1 = 2^2(3) - 1$$

$$23 = 24 - 1 = 8(3) - 1 = 2^3(3) - 1$$

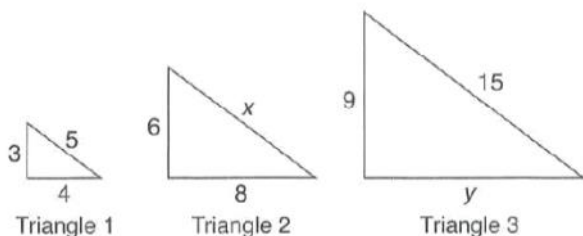
$$47 = 48 - 1 = 16(3) - 1 = 2^4(3) - 1$$

$$95 = 96 - 1 = 32(3) - 1 = 2^5(3) - 1$$

Rule: $2^n(3) - 1, n = 0, 1, 2, \dots$

FINDING A PATTERN WITH SIMILAR FIGURES

Example The triangles are similar. Find the measures of the missing sides.



	Vertical Leg	Horizontal Leg	Hypotenuse
Triangle 1	3	4	5
Triangle 2	6	8	x
Triangle 3	9	y	15

Read down each column to find the pattern. You can quickly see that the sides of Triangles 2 and 3 are multiples of Triangle 1.

The missing side in Triangle 2 is **10**. The missing side in Triangle 3 is **12**.

This table shows the relationship between the length and width of several rectangles and their perimeters. Find the pattern and complete the table. Then solve problems 12–13.

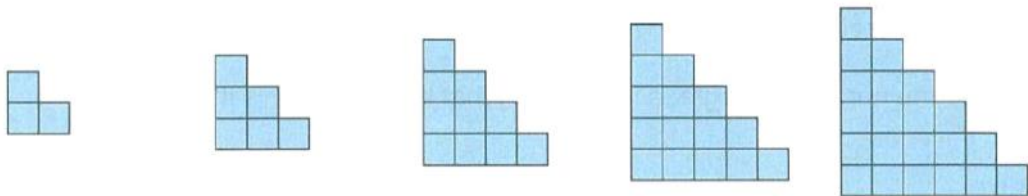
12. Find the perimeter of a rectangle with a length of 6 and a width of 5.
13. Find the length of a rectangle with a width of 7 and a perimeter of 30.

Rectangle	Length	Width	Perimeter
<i>ABCD</i>	2	1	6
<i>EFGH</i>	3	2	10
<i>IJKL</i>	4	3	14
<i>MNOP</i>	5	4	
<i>QRST</i>			
<i>UVWX</i>			

Indirection Measurement - More Examples

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The stair steps below are built with cubes. Count the number of cubes in each set to complete the chart. Then solve the problems.



Number of steps	2	3	4	5	6
Number of cubes	3	6			

14. How many cubes are needed to make a stair step with 8 steps?
15. **Explain** How many blocks would you expect to find on the bottom row of a stair step with 20 steps? How do you know your answer is correct?